



22CSC71- DEEP LEARNING							
Programme & Branch	B.E. - Computer Science and Engineering	Sem.	Category	L	T	P	Credit
Prerequisites	Nil	7	PC	3	0	2	4
Preamble	This course provides an overview Neural Networks and Deep learning techniques for solving real world problems.						
Unit - I	Neural Network						9
Neural Network-Building Intelligent Machines-Limits of Traditional Computer Programs-Mechanics of Machine Learning-Neuron-Linear Perceptrons as Neurons-Feed-Forward Neural Networks-Linear Neurons and Their Limitations-Sigmoid, Tanh, and ReLU Neurons Softmax Output Layers							
Training Feed-Forward Neural Networks:Gradient Descent-Delta Rule and Learning Rates-Gradient Descent with Sigmoidal Neurons-Backpropagation Algorithm-Stochastic and Minibatch Gradient Descent-Test Sets, Validation Sets, and Overfitting-Preventing Overfitting in Deep Neural Networks							
Unit - II	Learning in Neural Network						9
Challenges with Gradient Descent- Local Minima in the Error Surfaces of Deep Networks - Model Identifiability - Local Minima in Deep Networks - Flat Regions in the Error Surface - Gradient Points in the Wrong Direction - Second-Order Methods -Momentum-Based Optimization-Learning Rate Adaptation							
Unit - III	Convolutional Neural Networks						9
Convolutional Neural Networks-Neurons in Human Vision-Shortcomings of Feature Selection-Vanilla Deep Neural Networks-Filters and Feature Maps-Description of the Convolutional Layer-Max Pooling-Architectural Description of Convolution Networks-Accelerating Training with Batch Normalization- Visualizing Learning in Convolutional Networks-Convolutional Filters to Replicate Artistic Styles-Convolutional Filters for Other Problem Domains							
Unit - IV	Autoencoders and Recurrent Neural Networks						9
Embedding and Representation Learning-Principal Component Analysis-Autoencoder Architecture-Denoising to Force Robust Representations-Sparsity in Autoencoders- Models for Sequence Analysis- Tackling seq2seq with Neural N-Grams- Implementing a Part-of-Speech Tagger- Dependency Parsing and SyntaxNet- Beam Search and Global Normalization- Stateful Deep Learning Models- Recurrent Neural Networks- Challenges with Vanishing Gradients- Long Short-Term Memory (LSTM) Units- Augmenting Recurrent Networks with Attention- Dissecting a Neural Translation Network							
Unit - V	Case studies- Applications						9
Introduction to Keras and Tensorflow- Building a CNN for image classification - Sentiment Analysis Model using LSTM – Deep Learning for Network intrusion detection – Speech recognition using Deep learning							
LIST OF EXPERIMENTS / EXERCISES:							
1.	Exploration of deep learning frameworks Tensorflow and Keras						
2.	Create a simple Neural network to perform classification						
3.	Test the performance of multi-layer neural network with different activation functions						
4.	Improve the performance of the neural network with hyper parameter tuning						
5.	Implement a Convolutional Neural Network model for image classification						
6.	Improve performance of Convolutional Neural Network by tuning hyper parameters						
7.	Implement an Auto encoder for dimensionality reduction						
8.	Implement Object detection using Convolution Neural Network						
9.	Develop a sentiment analysis model using LSTM						
10.	Improve performance of LSTM by tuning hyper parameters						
Lecture:45, Practical:30, Total:75							
TEXT BOOK:							
1.	Nikhil Buduma , Fundamentals of Deep Learning: Designing Next-Generation Machine Intelligence Algorithms, 1st Edition, O'Reilly Series, June 2017						
REFERENCES/ MANUAL / SOFTWARE:							
1.	Ian Goodfellow, Yoshua Bengio, and Aaron Courvill, "Deep Learning", MIT Press, 1st Edition, 2016.						
2.	Josh Patterson and Adam Gibson, "Deep Learning – A Practitioner"s Approach", 1 Edition, O'Reilly Series, 2017.						
3.	Indra den Bakker, "Python Deep Learning Cookbook", 1 Edition, Packt Publishing, 2017.						



COURSE OUTCOMES: On completion of the course, the students will be able to		BT Mapped (Highest Level)
CO1	apply the concepts of neural network to solve simple problems	Applying (K3) Precision(S3)
CO2	Utilize different approaches to improve learning in neural networks	Applying (K3), Precision(S3)
CO3	exemplify the concepts of CNN models and apply it for solving computer vision related problems	Applying (K3) Precision(S3)
CO4	Make use of autoencoders for dimensionality reductions and Apply the concepts of RNN models for solving sequential modeling problems	Applying (K3), Precision(S3)
CO5	Identify suitable deep learning models for developing real world applications	Applying (K3), Precision(S3)

Mapping of COs with POs and PSOs

COs/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1										3	1
CO2	3	2	1										3	1
CO3	3	2	1	1	1				1	1		1	3	1
CO4	3	2	1	1	1				1	1		1	3	1
CO5	3	2	1	1	1				1	1		1	3	1

1 – Slight, 2 – Moderate, 3 – Substantial, BT- Bloom's Taxonomy

ASSESSMENT PATTERN - THEORY

Test / Bloom's Category*	Remembering (K1) %	Understanding (K2) %	Applying (K3) %	Analyzing (K4) %	Evaluating (K5) %	Creating (K6) %	Total %
CAT1		50	50				100
CAT2		40	60				100
CAT3		40	60				100
ESE		40	60				100

* ±3% may be varied (CAT 1,2,3 – 50 marks & ESE – 100 marks)